

Module 1

Introduction to Communication Systems and Linear Signal Processing



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F311CommSys25

L1: Objectives of Today Lecture

- Overview of the course
- Evaluation components
- Introduction to communication systems

Evaluation Components

Component	Duration	Marks (300)	Date & Time	Nature of component
Quizzes	25 min	15% (45)	6 Quizzes (each of 9 Marks): 2 Announced, 4 Surprise, 1 Buffer	Close Book
Weekly Labs	2 hours	8% (24)	Continuous	Close Book
Project	-	6% (18)	Continuous	Open Book
Lab Test	TBA	6 % (18)	End of Semester (TBA)	Close
Mid-Semester Test	As per AUGSD	25% (75)	As per AUGSD	Close Book
Comprehensive Exam	As per AUGSD	40% (120)	As per AUGSD	Open & Close Book

**Announce quizzed will be informed during Friday Lecture.*

- The weekly lab series is structured to guide students through the progressive development of a modular, end-to-end communication system simulator.
- Software: Python
- The simulator emphasizes the use of data structures and object-oriented programming to model each communication block.
- Students can also perform standalone validation of individual modules using MATLAB.
- Subsequent lab tasks will make extensive use of previous lab tasks.
- Two experiments on Emona hardware kits.

Lab Project: 6%

- Group of two students from the same section.
- Software: Python
- Objective: Demo of E2E simulator for both real-time audio and real-time text transmission
- Deadline: November, 2025

- Based on weekly tasks: close book.

- M1: Introduction to Communication Systems and Linear Signal Processing
- M2: Analog modulation and demodulation
- M3: Digital Signal Representation and Transmission
- M4: Noise in Communication Systems
- M5: Digital Modulations
- M6: Information Theory and Channel Coding
- M7: Concluding Insights and Recent Trends

- TB1: B.P. Lathi and Zhi Ding, “Modern Digital and Analog Communication Systems”, 4th Edition, Oxford University Press, 2017, South Asia Edition (Adapted by H.M. Gupta)
- R1: John G. Proakis, Masoud Salehi, “Communication Systems Engineering”, 2nd Edition, Pearson, 2002.

Human Speech and Hearing System

- Human voice: Transmits sound (longitudinal waves) signals with a finite power at a speed of 1225km/h.
- Speech frequency: 300Hz to 3.300 kHz
- Hearing frequency: 20 Hz to 20 kHz
- Signal loses power with the distance $P_r \propto \frac{P_t}{d^\alpha}$, $\alpha \geq 1$
- Human ear is a highly sensitive receiver: Receives signal with an intensity as low as $10^{-12} \text{ W/m}^2 (=0) \text{ dB}$.

Human Speech for Long Distance Communication

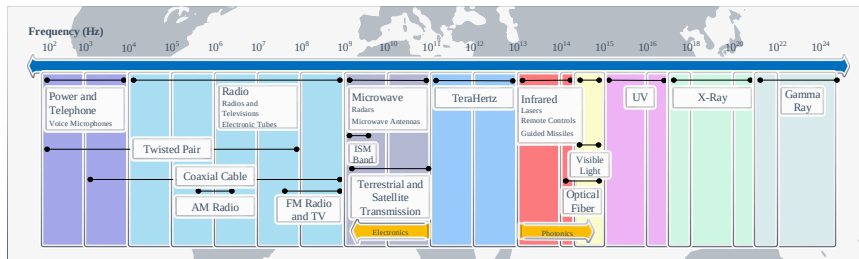
- Delay (speed of sound wave is low)
- Power (limited transmit power)
- Interference (same frequency used by all of us)

Alternative to Sound Wave?

Ideal characteristics for that wave or carrier or transportation system:

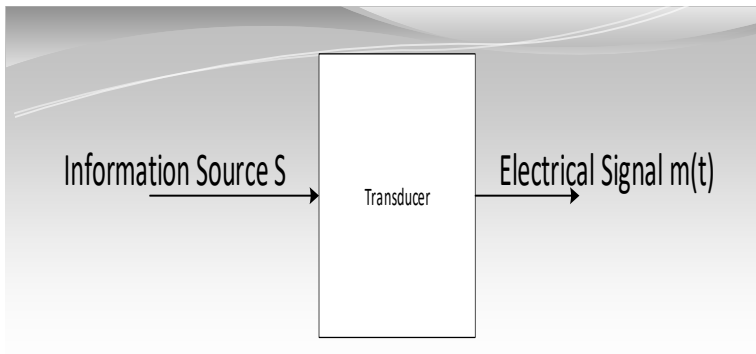
- Propagates with the highest possible speed
- Has enormous possibility of different frequencies (important for waves)

Electromagnetic Spectrum



- 2G (800, 1800 MHz), 3G, 4G: sub-GHz bands.
- We are moving from low frequency to higher frequency usage.
- 5G uses less than 50 GHz.
- 6G might use upto 1000 GHz.

- Conversion of physical information to electrical signal

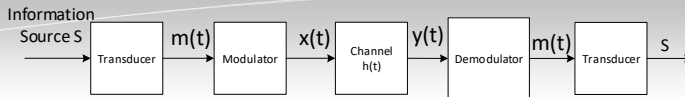


EM Wave Propagation

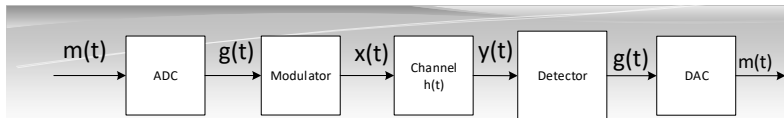
- Conversion of electrical signal to EM wave
- Guided medium or wireline (transmission lines, coaxial cables, fiber optics)
- Unguided medium or wireless

- Analog transmission
- Digital transmission

Electronic Communication Systems: Analog

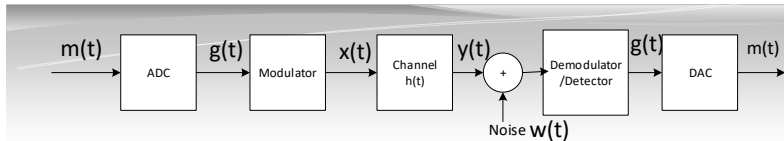


Electronic Communication Systems: Digital



Additive Noise

- More practical scenario



- Model for $w(t)$: Random

L2: Objectives of Today Lecture

- Signal Modeling
- Fourier Transform
- Energy, Power, Bandwidth

- Audio
- Text
- Image
- Video

- Infinite-Duration:

$$m(t) = \sum_{k=1}^{30} \frac{1}{k} \cos(2\pi k f_0 t).$$

- Finite-Duration t_0

$$m_w(t) = m(t)w(t), \quad \text{where} \quad w(t) = \begin{cases} 1, & 0 \leq t \leq t_0, \\ 0, & \text{otherwise.} \end{cases}$$

Signal Modeling of Information Source: Text

- Text: Character to bits (ASCII Coding)
- Binary to waveform
- For physical transmission over a channel, each bit—'1' or '0'—must be represented by a corresponding waveform.
- Signal modeling: Which waveform for bit '1' and which waveform for bit '0'
- Sinusoidal, Sinc, rectangular...
- What should be the criteria?

Important Quantities for Communications

- Power and energy of signals
- Bandwidth of the signal

Spectral Content: ω vs f

Fourier Transform

- $M(\omega) = \int_{-\infty}^{\infty} m(t)e^{-j\omega t}dt$
- $M(f) = \int_{-\infty}^{\infty} m(t)e^{-j2\pi ft}dt$
- $m(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} M(f)e^{+j\omega t}d\omega$
- $m(t) = \int_{-\infty}^{\infty} M(f)e^{+j2\pi ft}df$
- Condition on $m(t)$: deterministic

Fourier Transform Table (BP Lathi)

$g(t)$	$G(f)$	
1 $e^{-at}u(t)$	$\frac{1}{a + j2\pi f}$	$a > 0$
2 $e^{at}u(-t)$	$\frac{1}{a - j2\pi f}$	$a > 0$
3 $e^{-a t }$	$\frac{2a}{a^2 + (2\pi f)^2}$	$a > 0$
4 $te^{-at}u(t)$	$\frac{1}{(a + j2\pi f)^2}$	$a > 0$
5 $t^n e^{-at}u(t)$	$\frac{n!}{(a + j2\pi f)^{n+1}}$	$a > 0$
6 $\delta(t)$	1	
7 1	$\delta(f)$	
8 $e^{j2\pi f_0 t}$	$\delta(f - f_0)$	
9 $\cos 2\pi f_0 t$	$0.5[\delta(f + f_0) + \delta(f - f_0)]$	
10 $\sin 2\pi f_0 t$	$j0.5[\delta(f + f_0) - \delta(f - f_0)]$	
11 $u(t)$	$\frac{1}{j2\pi f} + \frac{1}{2}\delta(f)$	
12 $\text{sgn } t$	$\frac{2}{j2\pi f}$	
13 $\cos 2\pi f_0 t u(t)$	$\frac{1}{4}[\delta(f - f_0) + \delta(f + f_0)] + \frac{j2\pi f}{(2\pi f_0)^2 - (2\pi f)^2}$	
14 $\sin 2\pi f_0 t u(t)$	$\frac{1}{4j}[\delta(f - f_0) - \delta(f + f_0)] + \frac{2\pi f_0}{(2\pi f_0)^2 - (2\pi f)^2}$	
15 $e^{-at} \sin 2\pi f_0 t u(t)$	$\frac{2\pi f_0}{(a + j2\pi f)^2 + 4\pi^2 f_0^2}$	$a > 0$
16 $e^{-at} \cos 2\pi f_0 t u(t)$	$\frac{a + j2\pi f}{(a + j2\pi f)^2 + 4\pi^2 f_0^2}$	$a > 0$
17 $\Pi\left(\frac{t}{\tau}\right)$	$\tau \text{sinc}(\pi f \tau)$	
18 $2B \text{sinc}(2\pi Bt)$	$\Pi\left(\frac{f}{2B}\right)$	
19 $\Delta\left(\frac{t}{\tau}\right)$	$\frac{\tau}{2} \text{sinc}^2\left(\frac{\pi f \tau}{2}\right)$	
20 $B \text{sinc}^2(\pi Bt)$	$\Delta\left(\frac{f}{2B}\right)$	
21 $\sum_{n=-\infty}^{\infty} \delta(t - nT)$	$f_0 \sum_{n=-\infty}^{\infty} \delta(f - nf_0)$	$f_0 = \frac{1}{T}$
22 $e^{-t^2/2\sigma^2}$	$\sigma\sqrt{2\pi} e^{-2(\sigma\pi f)^2}$	

Fourier Transform Properties (BP Lathi)

Properties of Fourier Transform Operations

Operation	$g(t)$	$G(f)$
Superposition	$g_1(t) + g_2(t)$	$G_1(f) + G_2(f)$
Scalar multiplication	$kg(t)$	$kG(f)$
Duality	$G(t)$	$g(-f)$
Time scaling	$g(at)$	$\frac{1}{ a } G\left(\frac{f}{a}\right)$
Time shifting	$g(t - t_0)$	$G(f)e^{-j2\pi ft_0}$
Frequency shifting	$g(t)e^{j2\pi ft_0}$	$G(f - f_0)$
Time convolution	$g_1(t) * g_2(t)$	$G_1(f)G_2(f)$
Frequency convolution	$g_1(t)g_2(t)$	$G_1(f) * G_2(f)$
Time differentiation	$\frac{d^n g(t)}{dt^n}$	$(j2\pi f)^n G(f)$
Time integration	$\int_{-\infty}^t g(x) dx$	$\frac{G(f)}{j2\pi f} + \frac{1}{2} G(0)\delta(f)$

Fourier Transform: Conjugate Symmetry Property

- For real signals: $G(-f) = G^*(f)$
- Real signals: Amplitude spectrum $|G(-f)| = |G(f)|$ and Phase spectrum $\phi(-f) = -\phi(f)$

Parseval's Theorem

$$\int_{-\infty}^{\infty} |x(t)|^2 dt = \int_{-\infty}^{\infty} |X(f)|^2 df$$

Convolution Theorem

$$m(t)g(t) \leftrightarrow M(f) * G(f)$$

$$m(t) * g(t) \leftrightarrow M(f)G(f)$$

Convolution and Multiplication

Duration of $m(t)$: T_m .

Duration of $g(t)$: T_g

$$y(t) = m(t)g(t)$$

Duration of $y(t)$: $\min(T_m, T_g)$

Bandwidth of $M(f)$: B_m .

Bandwidth of $G(f)$: B_g

$$Y(f) = M(f) * G(f)$$

Bandwidth of $Y(f)$: $B_m + B_g$

Energy of Signal

- SaS $E_m = \int_{-\infty}^{\infty} |m(t)|^2 dt$
- CommSys $E_m = \int_0^T |m(t)|^2 dt$

Average Power of Signal

- Average power: $P_m = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} |m(t)|^2 dt$
- CommSys

$$P_m = \frac{1}{T_0} \int_0^{T_0} |m(t)|^2 dt.$$

Total Energy E and Average Energy per Bit E_b

A digit bit '1' can be represented by a rectangular pulse of width 1 second and the digit bit '0' can be by a negative rectangular pulse of width 1 second. Compute total energy E and average energy per bit E_b of (a) a digital '1' bit (b) a randomly generated finite digital bit-stream (c) a randomly generated infinite digital bit-stream.

Average Power P_{av} and Average Power per Bit P_b

A digit bit '1' can be represented by a rectangular pulse of width 1 second and the digit bit '0' can be by a negative rectangular pulse of width 1 second. Compute average power per bit P_b of (a) a digital '1' bit (b) a randomly generated finite digital bit-stream (c) a randomly generated infinite digital bit-stream. Assume amplitude 1 and equi-probable transmission of bit '1' and bit '0'

A digit bit '1' can be represented by a rectangular pulse of width 1 second and the digit bit '0' can be represented with no signal of pulse of width 1 second. Compute average energy per bit E_b .

Rectangular and Sinc Functions in Signal Processing

- $\text{sinc}(\pi x) = \frac{\sin(\pi x)}{\pi x}$
- $\Pi\left(\frac{x}{2a}\right)$

FT of Rectangular Pulse

- $g(t) = \Pi(\frac{t}{T})$
- FT?
- $G(f) = T \text{sinc}(\pi f T)$
- Plot $g(t)$ and $G(f)$
- What is signal BW?

FT of Sinc Pulse

- $g(t) = 2B\text{sinc}(2\pi Bt)$
- FT?
- $G(f) = \Pi(\frac{f}{2B})$
- Plot $g(t)$ and $G(f)$
- What is signal BW?

CW Problem

Energy of signal $m(t) = e^{-2t}u(t)$?

- Energy of signal $m(t) = 100\text{sinc}(200\pi t)$?

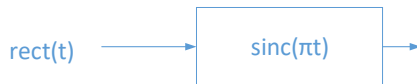
- Find the bandwidth of the signal $m(t) = 100\text{sinc}(200\pi t)$?

Signal Bandwidth

One of the definitions: Range of frequency where the signal power/energy is non-zero.

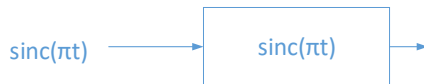
CW Problem

Find the BW of signal at the output:



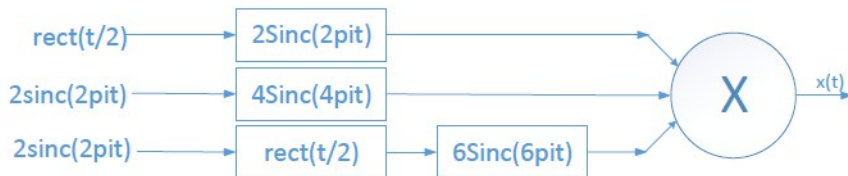
CW Problem

Find the energy of the signal at the output:



CW Problem

Find the BW of signal at the output:



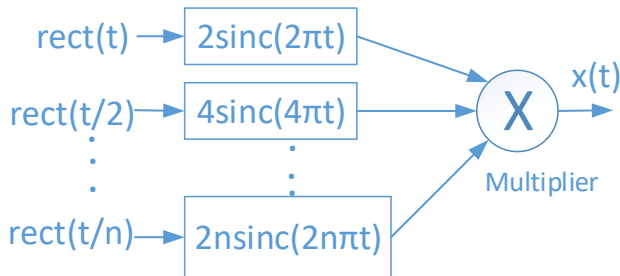
Ans: 3Hz

L3: Objectives of Today Lecture

- Few numerical problems on previous lecture
- Distortion-less transmission
- Energy and Power Spectral Density
- Auto-correlation function

CW Problem

N rectangular pulses with pulse width of 1 to n , where $n > 0$ are shaped by filters $2\text{sinc}(2\pi t)$ to $2n\text{sinc}(2n\pi t)$, respectively, and fed to a common multiplier. Find the bandwidth of the resultant signal $x(t)$.



CW Problem

A signal $m(t) = 2000 \operatorname{sinc}(2000\pi t)$ is passed through a circuit which produces an output signal $m^n(t)$, where n is a positive integer. For example, for $n = 2$, the circuit is called a squaring device. Calculate the bandwidth of the output signal.

Signals $m_1(t) = 10^4 \Pi(10^4 t)$ and $m_2(t) = \delta(t)$ are applied at the inputs of ideal low pass filters $F_1(f) = \Pi\left(\frac{f}{20000}\right)$ and $F_2(f) = \Pi\left(\frac{f}{10000}\right)$, respectively. The outputs $x_1(t)$ and $x_2(t)$ of these filters are multiplied to obtain the signal $x(t) = x_1(t)x_2(t)$. Find the bandwidth of signal $x(t)$.

Distortionless Transmission

Ideal

$$m(t) * h(t) = m(t)$$

Distortionless channel

$$|H(f)| = k$$
$$\phi(f) = -2\pi f t_d$$

A distortionless channel multiplies each component by the same factor and delays each component by the same amount of time.

Distortion-less Or Distorted

- Assume $m(t)$ is transmitted over channel

Channel Type 1:

$$H(f) = ke^{-j2\pi ft_d}, |f| < B$$

$$H(f) = 0, |f| > B$$

- $y(t) = km(t - t_d)$

Channel Type 2:

$$H(f) = (1 + k \cos 2\pi fT)e^{-j2\pi ft_d}, |f| < B$$

$$H(f) = 0, |f| > B$$

- $y(t) = m(t - t_d) + \frac{k}{2}[m(t - t_d - T) + m(t - t_d + T)]$

Distortion-less Or Distorted

- Assume $m(t)$ is transmitted over channel

$$|H(f)| = 1$$

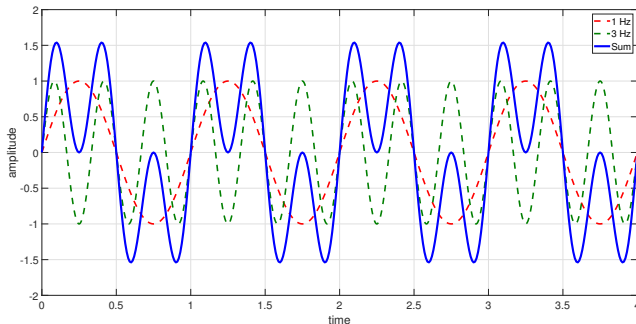
$$\theta(f) = -2\pi f t_d - k \sin(2\pi f T), k \ll 1$$

- Output signal

$$y(t) = m(t - t_d) + \frac{k}{2}m(t - t_d - T) - \frac{k}{2}m(t - t_d + T)]$$

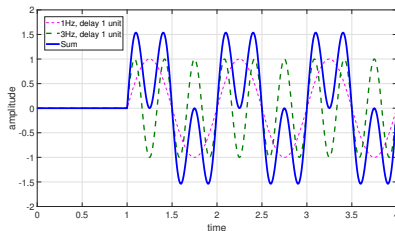
Input Signal: Distortion-less Or Distorted

$$m(t) = \sin(2\pi t) + \sin(6\pi t)$$

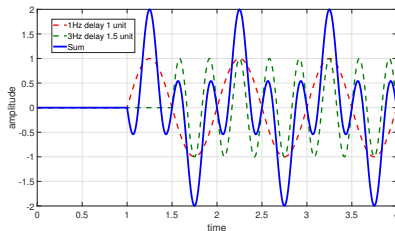


Output Signal: Distortion-less Or Distorted

Linear phase channel: Equal delay to each frequency component



Non-Linear phase channel: Unequal delay to each frequency component



Phase Difference for Distortion-less transmission

- Input signal

$m(t) = \sin(10\pi t) + \sin(20\pi t)$ is passed through a distortion-less channel to get the output signal

$$x(t) = \sin(10\pi t - \pi) + \sin(20\pi t + \phi)$$

Find the following:

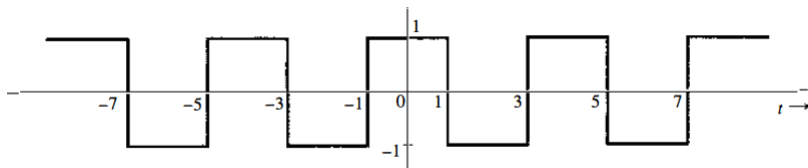
- Start time of input signal components
- Start time of output signal components
- Phase ϕ of the second signal component

L4: Objectives of Today Lecture

- Distortion-less transmission
- Energy and Power Spectral Density
- Auto-correlation function

Phase Difference and Distortion: Periodic Signal

- Input periodic signal $m(t)$



Explain the following:

- How to get a feel of phase of input periodic signal
- If the input period signal is passed through a distortion-less channel, how the phase of input signal changes?
- What is the relationship between the phase of output signal components?
- Answer lies in

$$m(t) = a_0 + \sum_{n=1}^{\infty} a_n \cos(2\pi n f_0 t) + \sum_{n=1}^{\infty} a_n \sin(2\pi n f_0 t)$$

Phase Difference and Distortion Non-periodic

- Input aperiodic signal: $m(t) = \Pi(t/T)$

Explain the following:

- How to get a feel of phase of input aperiodic signal
- If the input aperiodic signal is passed through a distortion-less channel, how the phase of input signal changes?
- What is the relationship between the phase of output signal components?
- Answer lies in $m(t) = \int_{-\infty}^{\infty} M(f)e^{j2\pi ft}df$

Distortion-less Transmission over Wireline Channels

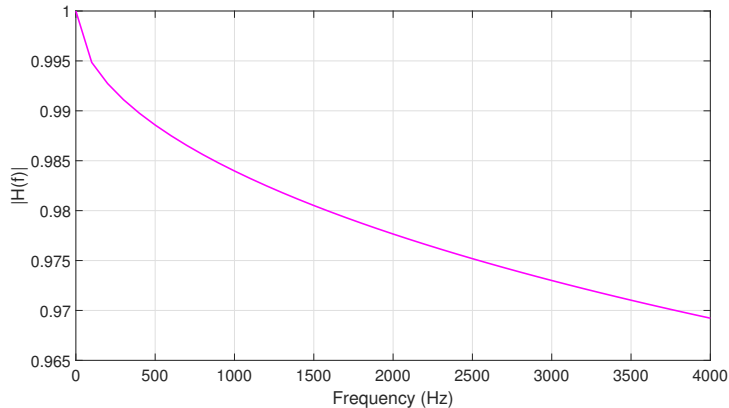
- Copper, CAT5, Optic Fiber, Coaxial Cables
- For copper: signal attenuation over a telephone line of length d :

$$H(f) = e^{-\gamma(f)d}$$

- $\gamma(f) = \alpha + j\beta = \sqrt{(R(f) + j2\pi fL(f))(G(f) + j2\pi fC(f))}$:
attenuation coefficient
- Low-Loss Line:

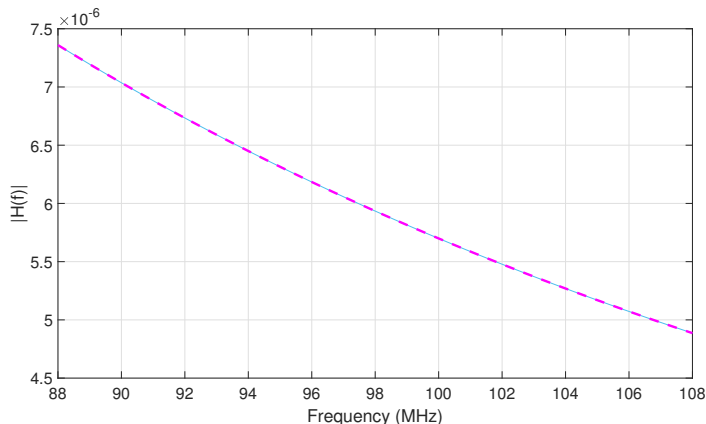
$$\alpha \approx \frac{1}{2}(R(f)\sqrt{\frac{C(f)}{L(f)}} + G(f)\sqrt{\frac{L(f)}{C(f)}})$$
$$\beta \approx 2\pi f\sqrt{L(f)C(f)}$$

Telephone Channel: Gain versus Frequency



Channel Gain in Wireless Medium: Free-Space

- Received power $|H(f)| = \sqrt{\frac{G_T G_R c^2}{(4\pi d)^2 f^2}}$
- G_T : Gain of transmitting antenna
- G_R : Gain of receiving antenna



Spectral Density

Variation of Signal Energy or Signal Power versus Frequency

Energy Spectral Density

- ESD $\psi(f) = |M(f)|^2$
- Energy $E_m = \int_{-\infty}^{\infty} \psi(f) df$

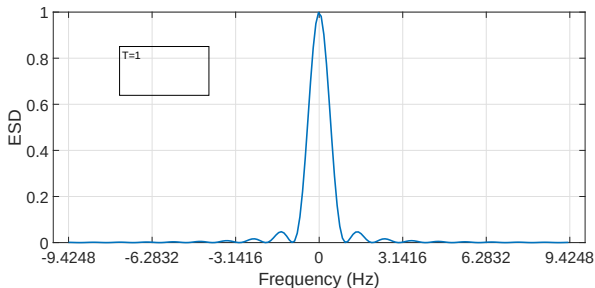
Plot Energy Spectral Density: Examples

- $m(t) = 2B \text{sinc}(2B\pi t)$
- $M(f) = \Pi(f/2B)$
- $m(t) = e^{-at}u(t)$
- $M(f) = \frac{1}{a+j2\pi f}$
- $m(t) = e^{-|t|}$
- $M(f) = \frac{2}{1+(2\pi f)^2}$
- $m(t) = \Pi(t/T)$
- $M(f) = T \text{sinc}(\pi fT)$
- $m(t) = \frac{2a}{t^2+a^2}$
- $M(f) = 2\pi e^{-2\pi a|f|}$

Estimate the BW from ESD.

Estimating Signal Bandwidth: Rectangular Pulse

- $m(t) = \Pi(t/T)$
- ESD $\psi(f) = |M(f)|^2 = T^2 \text{sinc}^2(\pi fT)$



Power Spectral Density of Signal

- PSD: $S_m(f) = \lim_{T \rightarrow \infty} \frac{|M_T(f)|^2}{T}$
- $P_m = \int_{-\infty}^{\infty} S_m(f) df$

L5: Objectives of Today Lecture

- Auto-correlation function
- Hilbert Transform

Auto-Correlation Function of Energy and Power Signals

Let $m(t)$ be an energy signal

- $R_m(\tau) = \int_{-\infty}^{\infty} m(t)m(t - \tau)dt$
- $R_m(\tau) = \int_{-\infty}^{\infty} m(t)m(t + \tau)dt$
- $E_m = R_m(0)$.

Let $m(t)$ be a power signal

- $R_m(\tau) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} m(t)m(t - \tau)dt$
- $R_m(\tau) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} m(t)m(t + \tau)dt$
- $P_m = R_m(0)$.

Properties of ACF

- $R_m(\tau) = R_m(-\tau)$
- $R_m(\tau)$ and $\psi_m(f)$ are FT pairs
- $R_m(\tau)$ and $S_m(f)$ are FT pairs

Proof: Fourier Transform of Auto-Correlation Function

- $\psi_m(f) = \int_{-\infty}^{\infty} R_m(\tau) e^{-j2\pi f\tau} d\tau$
- $\psi_m(f) = \int_{-\infty}^{\infty} \left[\int_{-\infty}^{\infty} m(t)m(t+\tau)dt \right] e^{-j2\pi f\tau} d\tau$
- $\psi_m(f) = \int_{-\infty}^{\infty} m(t) \left[\int_{-\infty}^{\infty} m(\tau+t) e^{-j2\pi f\tau} d\tau \right] dt$
- $\psi_m(f) = M(f) \int_{-\infty}^{\infty} m(t) e^{j2\pi ft} dt$
- $\psi_m(f) = M(f)M(-f) = |M(f)|^2 = \text{ESD}$
- Fourier transform of Auto-correlation function gives the ESD

Input-Output Relation

- $Y(f) = H(f)X(f)$
- $|Y(f)|^2 = |H(f)|^2|X(f)|^2$
- $S_y(f) = |H(f)|^2 S_x(f)$

CW Problems: Find Spectral Density

- $m(t) = A$
- $m(t) = A \cos 2\pi f_c t$
- $m(t) = A \cos(2\pi f_c t + \theta)$
- $M(f) = \text{sinc}(\pi f)$
- $M(f) = \frac{1}{a + j2\pi f}$

Tutorial-2 Problem

Find the auto-correlation function $R_x(\tau)$ and energy/power spectral density of the following signals

- $x(t) = e^{-at}u(t)$
- $x(t) = u(t)$

Solution

- $x(t) = e^{-\alpha t} u(t).$

$$X(f) = \frac{1}{\alpha + j2\pi f}$$

Energy spectral density

$$\mathcal{G}_X(f) = |X(f)|^2 = \frac{1}{\alpha^2 + 4\pi^2 f^2}.$$

Thus

$$R_X(\tau) = \mathcal{F}^{-1}[\mathcal{G}_X(f)] = \frac{1}{2\alpha} e^{-\alpha|\tau|}.$$

- $x(t) = u(t)$

$$R_X(\tau) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} u(t) u(t - \tau) dt = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{\tau}^{\frac{T}{2}} dt$$

$$= \lim_{T \rightarrow \infty} \frac{1}{T} \left(\frac{T}{2} - \tau \right) = \frac{1}{2}$$

$$S_X(f) = \mathcal{F}[R_X(\tau)] = \frac{1}{\omega} \delta(f).$$

Solution: Alternate for $u(t)$

- $x(t) = u(t)$

$$R_X(\tau) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} u(t) u(t + \tau) dt, \quad \tau > 0$$

Since $u(t) = 1$ for $t \geq 0$ and 0 otherwise:

$$R_X(\tau) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_0^{\frac{T}{2}} dt = \lim_{T \rightarrow \infty} \frac{\frac{T}{2}}{T} = \frac{1}{2}$$

Hence,

$$R_X(\tau) = \frac{1}{2}, \quad S_X(f) = \frac{1}{2} \delta(f)$$

3.8-1 Show that the autocorrelation function of $g(t) = C \cos(2\pi f_0 t + \theta_0)$ is given by $\mathcal{R}_g(\tau) = (C^2/2) \cos 2\pi f_0 \tau$, and the corresponding PSD is $S_g(f) = (C^2/4)[\delta(f - f_0) + \delta(f + f_0)]$. Hence, show that for a signal $y(t)$ given by

$$y(t) = C_0 + \sum_{n=1}^{\infty} C_n \cos(n2\pi f_0 t + \theta_n)$$

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the autocorrelation function and the PSD are given by

$$\mathcal{R}_y(\tau) = C_0^2 + \frac{1}{2} \sum_{n=1}^{\infty} C_n^2 \cos n2\pi f_0 \tau$$

$$S_y(f) = C_0^2 \delta(f) + \frac{1}{4} \sum_{n=1}^{\infty} C_n^2 [\delta(f - nf_0) + \delta(f + nf_0)]$$

Find the autocorrelation function of $m(t) \cos 2\pi f_c t$.

Solution

$$\begin{aligned} R_x(\tau) &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} x(t) x(t + \tau) dt \\ &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} m(t) m(t + \tau) \cos(2\pi f_c t) \cos(2\pi f_c(t + \tau)) dt \end{aligned}$$

Using the identity:

$$\cos A \cos B = \frac{1}{2} [\cos(A - B) + \cos(A + B)]$$

we get:

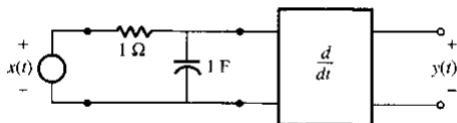
$$R_x(\tau) = \frac{1}{2} R_m(\tau) \cos(2\pi f_c \tau) + \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-T/2}^{T/2} m(t) m(t + \tau) \cos(4\pi f_c t + 2\pi f_c \tau) dt$$

The second term averages to zero, hence:

$$R_x(\tau) = \frac{1}{2} R_m(\tau) \cos(2\pi f_c \tau)$$

3.8-4 Find the mean square value (or power) of the output voltage $y(t)$ of the system shown in Fig. P3.8-4 if the input voltage PSD $S_x(f) = \Pi(\pi f)$. Calculate the power (mean square value) of the input signal $x(t)$.

Figure P3.8-4



$$\text{Input} = \frac{1}{\pi}$$

$$\text{Output} = \frac{1}{\pi} - \frac{1}{4}$$

Solution

$$S_x(f) = \Pi(\pi f)$$

$$P_x = \int_{-\infty}^{\infty} S_x(f) df = \int_{-1/2\pi}^{1/2\pi} df = \frac{1}{\pi}$$

RC part:

$$H_{RC}(f) = \frac{1}{1 + j2\pi f}$$

Differentiator:

$$H_{d/dt}(f) = j2\pi f$$

$$H(f) = H_{RC}(f) H_{d/dt}(f) = \frac{j2\pi f}{1 + j2\pi f}$$

$$H(f) = H_{RC}(f) H_{d/dt}(f) = \frac{j2\pi f}{1 + j2\pi f} = \frac{1}{1 - \frac{j}{2\pi x}}$$

$$|H(f)|^2 = \frac{\frac{1}{1}}{1 + \frac{1}{4\pi^2 x^2}}$$

$$S_y(f) = |H(f)|^2 S_x(f) = \frac{1}{1 + \frac{1}{4\pi^2 x^2}} \Pi(\pi f)$$

and

$$P_y = \int_{-\infty}^{\infty} S_y(f) df = \int_{-1/2\pi}^{1/2\pi} \frac{1}{1 + \frac{1}{4\pi^2 x^2}} df = f - \frac{\tan^{-1}(2\pi f)}{2\pi} \Big|_{-1/2\pi}^{1/2\pi} = \frac{1}{\pi} - \frac{1}{4}$$

L6: Objectives of Today Lecture

- Hilbert Transform
- Analytical Signal
- Band-pass Signal
- Low-pass Equivalent of of Band-pass Signal

Fourier Transform of $f(t) = \frac{1}{\pi t}$

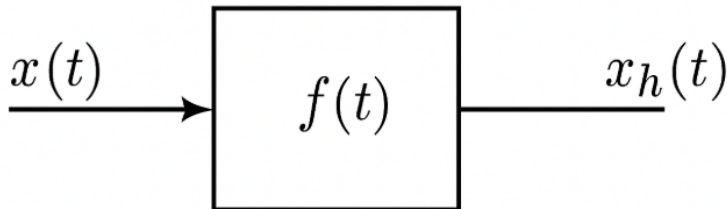
- Signum Function

$$\text{sgn}(t) = \begin{cases} +1, & \text{if } t > 0 \\ 0, & \text{if } t = 0 \\ -1, & \text{if } t < 0 \end{cases}$$

- $F(f) = FT[\frac{1}{\pi t}] = -j\text{sgn}(f)$

$$F(f) = \begin{cases} -j = 1 \cdot e^{-j\pi/2}, & f > 0 \\ 0, & f = 0; \\ j = 1 \cdot e^{j\pi/2}, & f < 0 \end{cases}$$

Hilbert Transform: Revisit



- $x_h(t) = x(t) * \frac{1}{\pi t} = \frac{1}{\pi} \int_{-\infty}^{\infty} x(\alpha) \frac{1}{t-\alpha} d\alpha$
- $X_h(f) = X(f)F(f) = -j\text{sgn}(f)X(f)$
- $|F(f)| = 1$ and phase $\phi(f) = -\pi/2$ for $f > 0$, $\phi(f) = +\pi/2$ for $f < 0$.
- Hilbert transform: changes phase of the input signal $x(t)$.

Hilbert Transform: Examples

- Show that Hilbert filter does not change the energy or power of the input signal.

Solution

- For power: use $|F(f)| = 1$ in PSD $S_y(f) = |F(f)|^2 S_x(f) = S_x(f)$
- For energy: use $|F(f)| = 1$ in ESD $\psi_y(f) = |F(f)|^2 \psi_x(f) = \psi_x(f)$

Hilbert Transform: Examples

- Hilbert transform of $x(t) = \cos(2\pi f_0 t)$
- $x_h = \cos(2\pi f_0 t - 90) = \sin(2\pi f_0 t)$
- Hilbert transform of $\sin(2\pi f_0 t) = \sin(2\pi f_0 t - 90) = -\cos(2\pi f_0 t)$

Hilbert Transform of $x(t) = \cos(2\pi f_0 t)$ using Frequency-Domain Approach

$$F(f) = -j \operatorname{sgn}(f).$$

$$x(t) = \cos(2\pi f_0 t), \quad X(f) = \frac{1}{2} [\delta(f - f_0) + \delta(f + f_0)].$$

$$X_h(f) = F(f)X(f) = \frac{1}{2} [(-j)\delta(f - f_0) + (+j)\delta(f + f_0)].$$

$$x_h(t) = \sin(2\pi f_0 t).$$

Hilbert Transform of $x(t) = \sin(2\pi f_0 t)$ using Frequency-Domain Approach

$$x(t) = \sin(2\pi f_0 t), \quad X(f) = \frac{1}{2j} [\delta(f - f_0) - \delta(f + f_0)].$$

$$X_h(f) = \frac{1}{2j} [(-j)\delta(f - f_0) - (+j)\delta(f + f_0)] = -\frac{1}{2} [\delta(f - f_0) + \delta(f + f_0)].$$

$$x_h(t) = -\cos(2\pi f_0 t).$$

Real-valued Signals: Negative Spectra

- $x(t) = \cos(2\pi f_0 t)$
- Fourier transform representation of $x(t)$
- $x(t) = \frac{1}{2} [e^{-j2\pi f_0 t} + e^{j2\pi f_0 t}]$
- $X(f) = \frac{1}{2} [\delta(f + f_0) + \delta(f - f_0)]$
- Real signals: both sided spectra
- $x(t) = e^{j2\pi f_0 t}$, $X(f) = \delta(f - f_0)$
- Complex-valued signals: one sided spectra

Analytic Signal

- Analytical signal: it is a complex-valued signal and contain one-sided frequency only. Define.

$$X_a(f) = \begin{cases} 2X(f), & f > 0 \\ 0, & f \leq 0; \end{cases}$$

- $X_a(f) = X(f) + \text{sgn}(f)X(f)$
- Show that analytical signal in time-domain: $x_a(t) = x(t) + jx_h(t)$.
- $x_a(t) = |x_a(t)|e^{-\tan^{-1} \frac{x_h(t)}{x(t)}}$
- $x_a(t)$: complex envelope of a signal $x(t)$.
- $|x_a(t)| = \sqrt{x^2(t) + jx_h^2(t)}$: envelope of $x(t)$

Analytic Signal: Example

- Find analytical signal of $x(t) = \cos(2\pi f_0 t)$.

Analytic Signal: Example

- Find analytical signal of $x(t) = \cos(2\pi f_1 t) + \sin(2\pi f_2 t)$.

Analytic Signal: Example

- Find $|x_a(t)|$ for $x(t) = \cos(200\pi t) + \cos(240\pi t)$.

$$x_a(t) = [\cos(200\pi t) + j \sin(200\pi t)] + [\cos(240\pi t) + j \sin(240\pi t)]$$

$$\Rightarrow x_a(t) = e^{j200\pi t} + e^{j240\pi t}$$

$$x_a(t) = e^{j220\pi t} (e^{-j20\pi t} + e^{+j20\pi t}) = 2e^{j220\pi t} \cos(20\pi t)$$

$$|x_a(t)| = 2 |\cos(20\pi t)|$$

The solution is correct.

The envelope of the analytical signal oscillates at the half of the difference frequency i.e., at a much lower frequency than the original signal.

Module 1 is concluded.